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Fórmula para el laplaciano en función de las derivadas de Wirtinger

Lema 1. *Sea* $f \in \mathcal{C}^2(\mathbb{D})$*, entonces* $\forall z \in \mathbb{D} : \Delta f(z) = 4\,\frac{\partial^2 f}{\partial z\,\partial \overline{z}}(z) = 4\,\frac{\partial^2 f}{\partial \overline{z}\,\partial z}(z)$.

***Demostración*:** Desarrollando las derivadas de Wirtinger en el lado derecho, tenemos:

$$\begin{aligned}
\frac{\partial^2 f}{\partial z\,\partial \overline{z}} &= \frac{\partial}{\partial z}\left(\frac{\partial f}{\partial \overline{z}}\right) = \frac{\partial}{\partial z}\left(\frac{1}{2}\left(\frac{\partial f}{\partial x} + i\,\frac{\partial f}{\partial y}\right)\right) \\
&= \frac{1}{2}\left[\frac{1}{2}\left(\frac{\partial^2 f}{\partial x^2} - i\,\frac{\partial^2 f}{\partial y\,\partial x}\right) + \frac{1}{2}i\left(\frac{\partial^2 f}{\partial x\,\partial y} - i\,\frac{\partial^2 f}{\partial y^2}\right)\right] \\
&= \frac{1}{4}\left(\frac{\partial^2 f}{\partial x^2} - i\,\frac{\partial^2 f}{\partial y\,\partial x} + i\,\frac{\partial^2 f}{\partial x\,\partial y} + \frac{\partial^2 f}{\partial y^2}\right) = \frac{1}{4}\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) = \frac{1}{4}\Delta f.
\end{aligned}$$

Por tanto, $\Delta = 4\,\frac{\partial^2}{\partial z\,\partial \overline{z}}$ (también se puede ver que $\Delta = 4\,\frac{\partial^2}{\partial \overline{z}\,\partial z}$ de forma análoga). ■

Referenciado en

- `Lem-laplaciano-cambio-variable`